Quarto, data frames, and basic plots

Overview

Quick review from last class

- R as a calculator, objects, data types
- Functions
- Vectors

More R

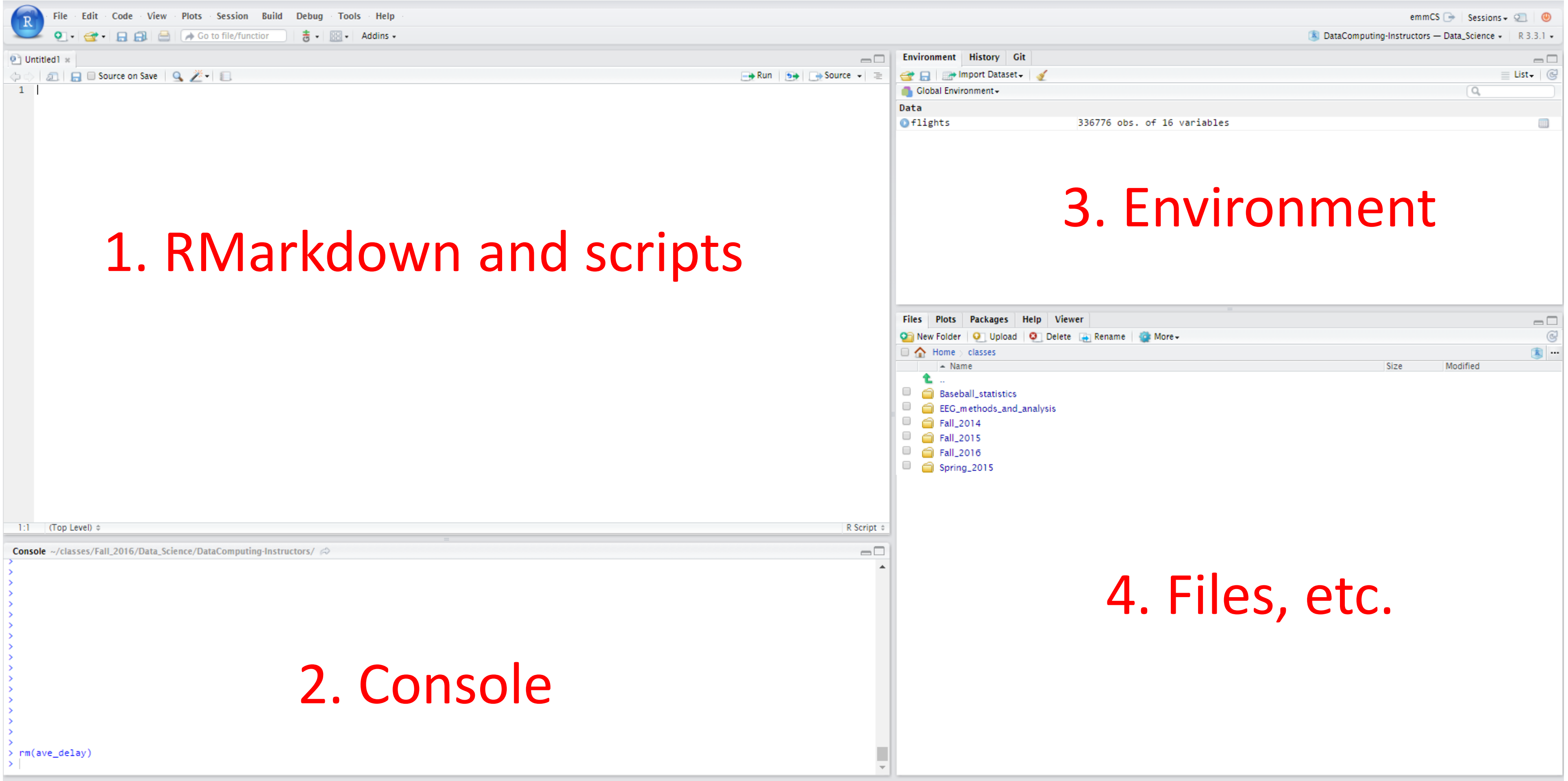
- R packages
- Quarto
- Data Frames

If there is time:

- Plots and statistics of categorical data
- Plots and statistics of quantitative data

Any questions about anything?

Please open up RStudio



Review: R Basics

Arithmetic:

```
> 2 + 2
> 7 * 5
```

Assignment of values to ***objects***:

```
> a <- 4
> b <- 7
> z <- a + b
> z
[1] 11
```

Review: Character strings and Booleans

```
a <- 7
s <- "s is a terrible name for an object"
b <- TRUE

class(a)
  [1] numeric

class(s)
  [1] character
```

Review: Functions

Functions use parenthesis: functionName(x)

```r
sqrt(49)
tolower("DATA is AWESOME!")
```

To get help

```r
? sqrt
```

One can add comments to your code

```r
sqrt(49)    # this takes the square root of 49
```

Review: Vectors

Vectors are ordered sequences of numbers or letters

The c() function is used to create vectors

```
v <- c(5, 232, 5, 543)
s <- c("statistics", "data", "science", "fun")
```

One can access elements of a vector using square brackets []

```
s[4]        # what will the answer be?
```

We can also apply functions to vectors

```
sum(v)
```

R packages

R packages

Packages add additional functionality to R

We will use many additional packages in this class

- ggplot2, dplyr, tidyr, etc.

There is a class specific package (SDS1110) I wrote that we will use to download homework and other files

SDS1110 package

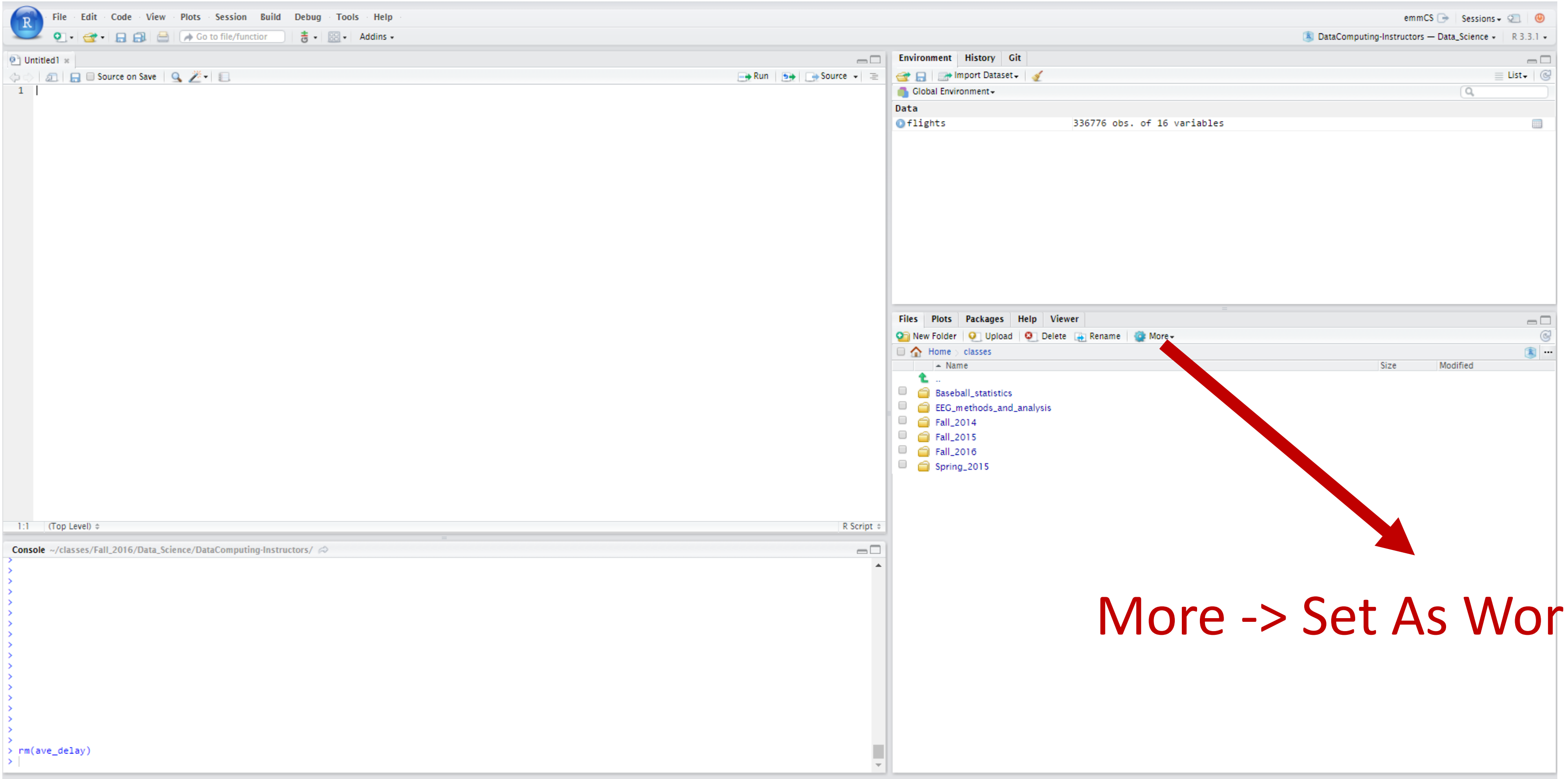
The SDS1110 package is already installed for you on the YCRC cluster

To use the functions in the package you can use

```
library(SDS1110)
download_homework(1)
```

Before downloading files, it is useful to create directories to store the data in and to set your working directory…

Setting your working directory



More -> Set As Working Directory

1. In the files tab, navigate to the directory that contains the .qmd file
2. Click More -> Set As Working Directory

Downloading class 2 code

We can use the SDS1110 package to get code for today's class by typing the following commands at the console:

```
library(SDS1110)

download_class_code(2)

download_class_code(2, TRUE)      # version with the answers
```

Questions?

Let's download homework 1 and the class 2 code now!

Quarto

Quarto

Quarto (.qmd files) allow you to embed written descriptions, R code, and the output of that code into a nice looking document

Makes it easy to do reproducible research!

Quarto

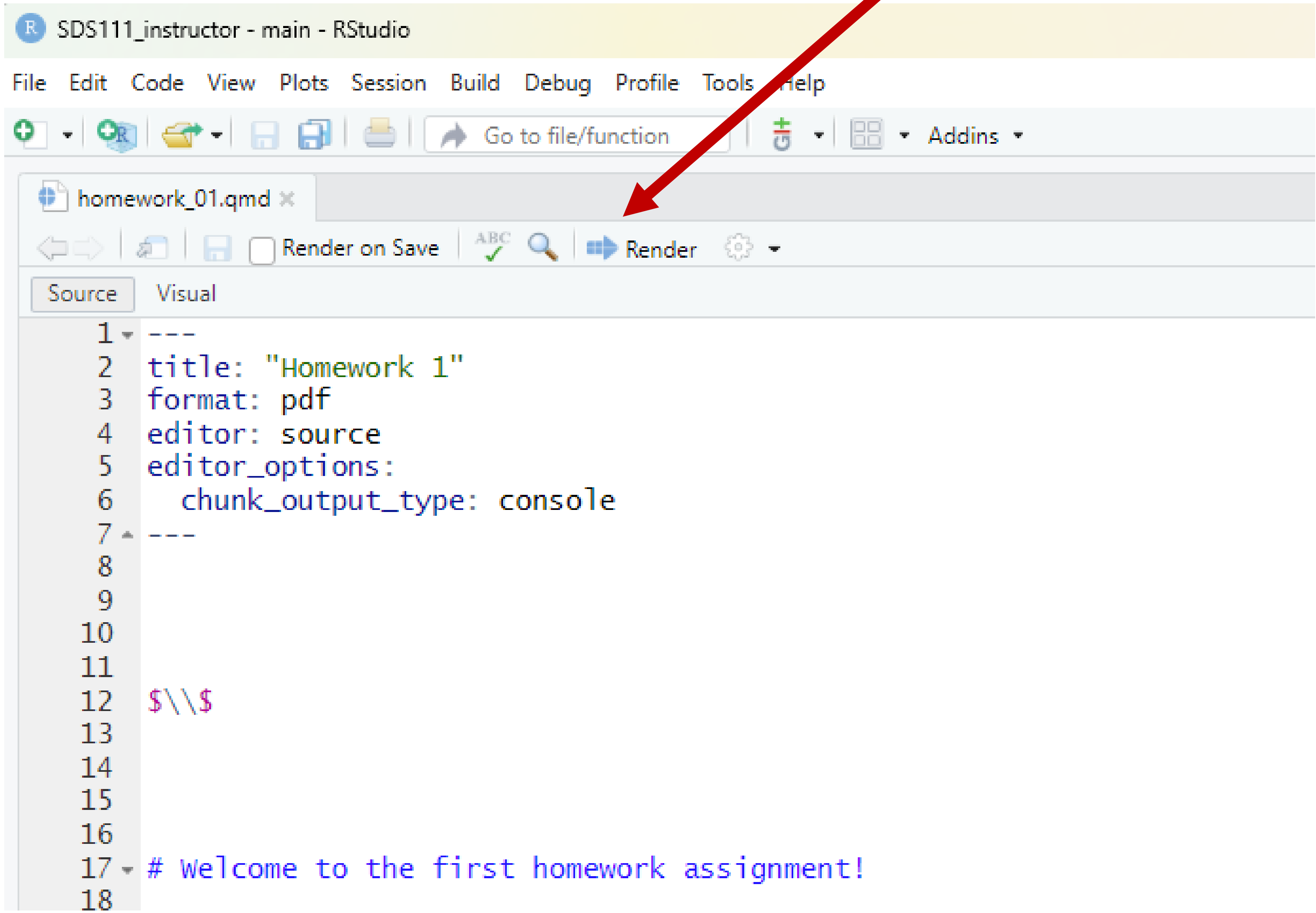
Everything in R chunks is executed as code:

````
```{r}
    # this is a comment
    # the following code will be executed
    2 + 3
```
````

Everything outside R chunks appears as text

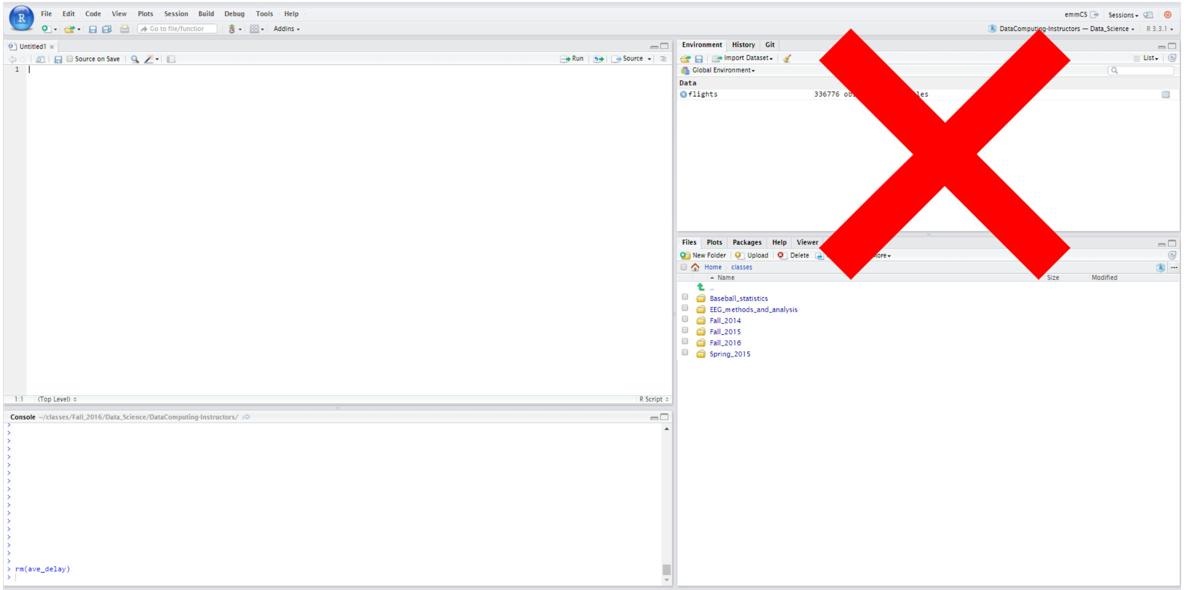
Render to a pdf

Turn in a pdf or html document with your solutions to Canvas



Quarto

Note: When you render a Quarto document, your Quarto document **does not have access to objects in the global environment**



Why is this a good thing???

Formatting in Quarto

We can add formatting to text outside the code chunks

Examples:

```
## Level 2 header
**bold**
![](image.png)
```

To avoid code that is hard to debug…

Only change a few lines at a time and then render your document to make sure everything is working!

If you document isn't rendering:

- **For code chunks:** use the # symbol to comment out code until you can find the line of code that is giving the error message
- **Outside of code chunk:** cut out part of the document until it renders and then paste it back

Announcement: Homework 1

Due Monday July 6th at 11pm

- I recommend getting started on this soon!
  - E.g., might be a good idea to work on it during office hours after this class…

To download the homework please do the following:

```
library(SDS1110)
download_homework(1)
```

From the file panel, open the homework and try knitting it

Announcement: Homework 1

Instructions for how to submit homework on Gradescope are on Canvas

- Please mark all pages that answers correspond to on Gradescope!

Be sure to also "show your work" by printing out any values you report

- Although don't print out hundreds of access pages of numbers

Ask/answer questions on Ed Discussions, but don't give away the solutions!

Questions?

Data frames

Data frames contain structured data

	age	body_type	diet	drinks	drugs	education
1	22	a little extra	strictly anything	socially	never	working on college/university
2	35	average	mostly other	often	sometimes	working on space camp
3	38	thin	anything	socially	NA	graduated from masters program
4	23	thin	vegetarian	socially	NA	working on college/university
5	29	athletic	NA	socially	never	graduated from college/university
6	29	average	mostly anything	socially	NA	graduated from college/university

Back to R: Data frames

Data frames contain structured data

```
library(SDS1110)
download_data("profiles_revised.csv")     #  only needs to be run once
profiles <- read.csv("profiles_revised.csv")
View(profiles)      # the View() function only works in RStudio!
```

	age	body_type	diet	drinks	drugs	education
1	22	a little extra	strictly anything	socially	never	working on college/university
2	35	average	mostly other	often	sometimes	working on space camp
3	38	thin	anything	socially	NA	graduated from masters program
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Data Frames

Variables

Cases

	age	body_type	diet	drinks	drugs	education
1	22	a little extra	strictly anything	socially	never	working on college/university
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Categorical and quantitative variables

Quantitative Variable

Categorical Variable

Cases (observational units)

	age	body_type	diet	drinks	drugs	education
1	22	a little extra	strictly anything	socially	never	working on college/university
2	35	average	mostly other	often	sometimes	working on space camp
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Data frames

We can extract the columns of a data frame as vector objects using the $ symbol

```
the_ages <- profiles$age
```

Can you get the sum of the ages of users in this data set?

```
sum(the_ages)
```

Let's try it in RStudio!

Questions?

Categorical data

Categorical variables

	age	body_type	diet	drinks	drugs	education
1	22	a little extra	strictly anything	socially	never	working on college/university
2	35	average	mostly other	often	sometimes	working on space camp
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What is a categorical variable?

Which variables in the profiles data frame are categorical?

How can we extract the drinks vector from the profiles data frame?

```
drinking_vec <- profiles$drinks
```

Categorical variables

For categorical variables, we usually want to view:

- How many items are each category   OR
- The proportion (or percentage) of items in each category

$$\text{Proportion in a category} = \frac{\text{number in that category}}{\text{total number}}$$

Calculating counts on a categorical variable

The count of how many items are in each category can be summarized in a ***frequency table***

	Desperately	**Not at all**	**Often**	**Rarely**	**Socially**	**Very often**		**Total**
Count	322	3267	5164	5957	41780	471		56961

In R we can create a frequency table using:

```
my_table <- table(data_vector)
```

Calculating proportions (relative frequencies)

We can convert a frequency table into a ***relative frequency table*** by dividing each cell by the total number of items

	Desperately	Not at all	Often	Rarely	Socially	Very often		Total
Count	0.006	0.057	0.091	0.105	0.733	0.008		1

In R we can create a relative frequency table using:

```
rel_freq_table <- prop.table(my_table)
```

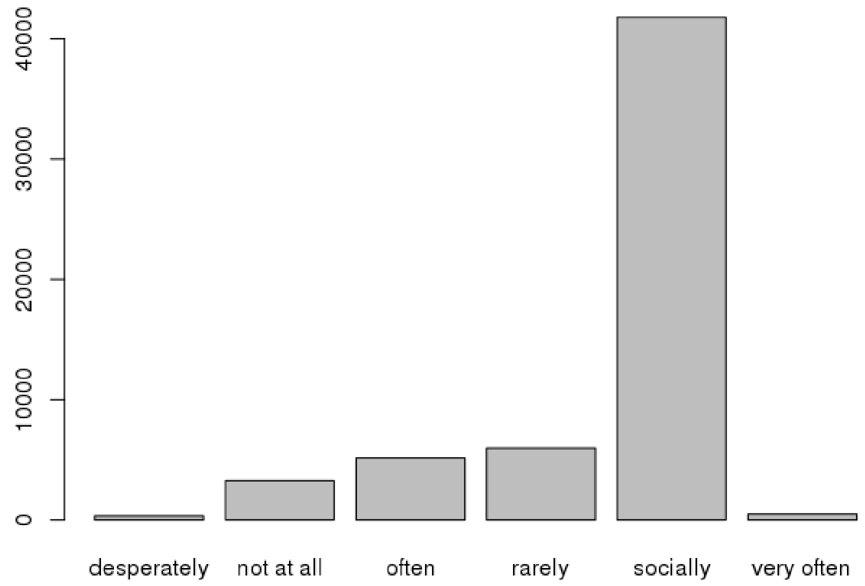
Note: this argument is a table not a vector!

Bar plots (pun intended?)

We can visualize the number of items in each category using a bar plot

barplot(my_table)

The height of each bar corresponds to how many items are in each category



Let's try it in RStudio!

Details matter!

Can you figure out how to label the axes?

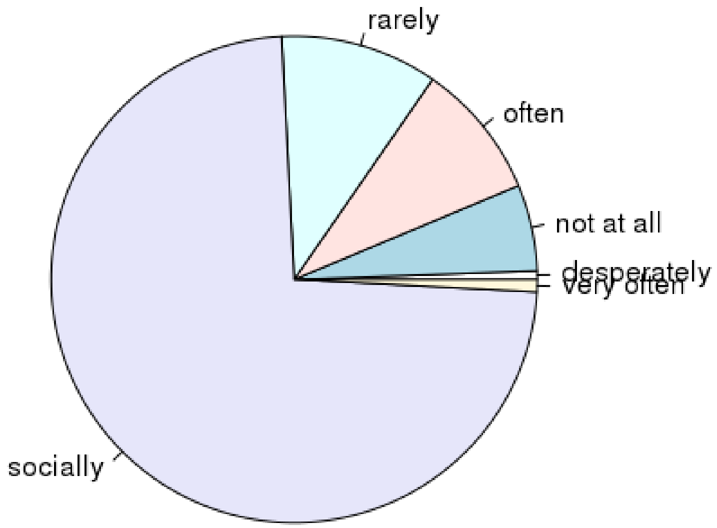
- A: ? barplot
- A: xlab and ylab!

```
barplot(drinks_table,
        ylab = "Count",
        xlab = "Type of drinker",
        main = "Counts of different types of drinkers")
```

Pie charts

We can also use the pie() function to create pie charts

```
pie(drinks_table)
```



Questions?

For next class…

Homework 1 is due on Gradescope by 11pm on Monday July 6^{th}

- Instructions for how to submit homework on Gradescope are on Canvas